

Dhruv H

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EDUCATION

B.M.S College of Engineering

Bachelor of Engineering in Information Science

- CGPA: 9.232/10.0

Base PU College

Pre-University Course (PUC)

- Percentage: 92.16%

V.V.S Sardar Patel High School

Secondary School Leaving Certificate (SSLC)

- Percentage: 99.36%

Bengaluru, Karnataka

Aug. 2023 – Present

Bengaluru, Karnataka

2021 – 2023

Bengaluru, Karnataka

2021

PROJECTS

dvara | *Python, FastAPI, Redis, Bloom Filter, PyPI* | [visit](#) ↗

- Published a PyPI package for malicious URL detection using a two stage Redis → PostgreSQL Bloom Filter pipeline, achieving 145K URLs/sec throughput.
- Compressed 268K malicious URLs into 5.14MB at 0.1% FPR; production deployed and publicly available on PyPI, Redis bit checks handle clean traffic entirely, eliminating DB load for benign URLs.

FluxGate | *React, FastAPI, Redis, PostgreSQL, Python*

- Built a distributed rate limiting service with 4 algorithms (token bucket, sliding window, leaky bucket) using Redis Lua atomic scripts to eliminate race conditions in concurrent environments.
- Load tested at 4K+ requests across 20 concurrent clients with per key enforcement; implemented hot config reload and real time WebSocket analytics dashboard showing live RPS and throttle metrics.

Exoplanet Atmospheric Retrieval | *Python, XGBoost, Scikit-learn, Optuna* | *Manuscript under review, Scientific Reports*

- Benchmarked MLP, SVR and XGBoost as lightweight alternatives to Bayesian atmospheric retrieval on the Ariel Big Data Challenge dataset, retrieving 5 gas abundances + equilibrium temperature from transmission spectra.
- Preprocessing (52 bin wavelength interpolation, log scaling, feature augmentation) proved the dominant performance factor; XGBoost generalized best to real JWST NIRSpec observations of WASP-39b, matching published results.

RAG Document Q&A | *Python, LangChain, FastAPI, ChromaDB, Ollama*

- Built a fully offline RAG pipeline — upload any PDF, query it in natural language; LangChain orchestrates chunk retrieval from ChromaDB and grounded answer generation via local llama3.2.
- Implemented a custom RAGAS-style eval harness with experiment tracking; identified chunk-formatting as root cause of retrieval failures through systematic ablation over top-k and chunk size.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, HTML/CSS

Frameworks/Libraries: FastAPI, Express.js, Node.js, React, Spring Boot, PyTorch, LangChain, Pandas, NumPy

Infrastructure & Cloud: Redis, PostgreSQL, MySQL, MongoDB, ChromaDB, Docker, Docker Compose, AWS

Developer Tools: Git, GitHub, VS Code, UNIX, Postman, Jupyter Notebook, Claude Code, Cursor

Concepts: Data Structures & Algorithms, Probabilistic Data Structures, Distributed Systems, OOP, REST APIs, Microservices, Machine Learning, Deep Learning, Graph Neural Networks, Retrieval Augmented Generation (RAG), Vector Embeddings, WebSockets, Lua, Competitive Programming (Leetcode/Hackerrank)

CERTIFICATIONS

Nutanix Certified Associate (NCA)

- Certified in hyperconverged infrastructure, cloud computing fundamentals, and Nutanix AOS administration.

JPMorgan Chase: Software Engineering Job Simulation (Forge)

- Built Kafka integrated Spring Boot microservice with JPA persistence, REST endpoints, and Maven based testing.

NVIDIA Generative AI Explained

- Covered fundamentals of generative AI including GANs, transformers, and their practical applications in modern software systems.

ADDITIONAL SKILLS

Soft Skills: Collaborative Team Player, Analytical Problem-Solving, Adaptability, Critical Thinking, Innovation, Effective Communication

Languages: English (Fluent), Hindi (Proficient), Kannada (Proficient), Gujarati (Conversational)